

Exam Date & Time: 27-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

Programming in C sharp [CSE-4307]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With a neat diagram explain common language runtime. (3)
 - A)
 - B) Apply your coding knowledge to resolve scope clash and with an example program show how the new variable named `j` that we declare in the `Main()` method *hides* the class-level variable with the same name. (3)
 - C) Explain advantages of managed code in detail. (4)
- 2) Illustrate the concept of break in a C# program. (2)
 - A)
 - B) What is enumeration? Illustrate this concept with C# program to show how values are assigned for months in a year. (5)
 - C) Demonstrate interface inheritance through a C# program and show how a class is inheriting only one interface. (3)
- 3) Illustrate how binary operators (-) inside a class Complex can be overloaded in a C# program. (3)
 - A)
 - B) Choose "addnum" and "subnum" as two delegate names in C# program and show how two values are added and subtracted by invoking delegates. (4)
 - C) Consider C# reflection and use `typeof` operator to get assembly of variable `t` using the `Assembly` property (3)
- 4) Demonstrate `GC.GetGeneration()` Method and `GC.GetTotalMemory()` Method through C# program. (3)
 - A)
 - B) Consider Dictionary < TKey, TValue > class and demonstrate adding key/value pairs in (5)

the Dictionary by with and without Using Add() method in C# Program.

- C) Show different requirements for an assembly to be CLS compliant (2)
- 5) Imagine you're using a Microsoft application, and you attempt to use a feature that you have not used before, illustrate with a diagram what an assembly's code group membership might look like with code from the Internet published by a named organization that has signed the assembly with a certificate. (3)
 - A)
 - B) Explain how .NET provides three levels of code groups that are independently managed and exist in parallel. (3)
 - C) Demonstrate how WCF supports the development of services following the principles of service-oriented architecture with key features. (4)

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